Alphabet Inc.

Massive Capex From a Position of Strength; Reiterate Overweight & $395 PT

Google reported very strong 4Q results with accelerating growth in Search and Cloud, but capex will be the story coming out. The 2026 capex outlook for $175B-$185B was well above our previous (Street high) estimate of $142B. **Importantly, we believe the doubling of capex Y/Y comes from a position of strength, with Gemini now at 750M MAUs & showing significantly higher engagement per user, Google Cloud revenue accelerating to +48% w/backlog up 55% Q/Q to $240B, & Search revenue accelerating to +17% as AI continues to expand the market.** Pushback will come from Google becoming increasingly capital intensive and pressuring both operating income and FCF. Depreciation will accelerate meaningfully this year (we model +73%), but Google also emphasized that it will continue to drive efficiencies across the business in tech/eng, server deployment, TPU improvements, & headcount, among other areas. The significant step-up in capex will weigh on FCF—we model $29B this year (-61%)—but Google currently has a net cash balance of $80B. While some might say the financial profile and outsized spending shifts Google closer to Meta, we believe we are seeing clear returns on Google's investments across Gemini, Cloud, and Search, and Google differentiates with very meaningful multi-year backlog. Coming out of earnings, our 2026/2027 revenue increases 2-4%, our operating income increases 3-6%, but our FCF comes down 45-55%. Our GAAP EPS increases 2-4%, and we now project $13.60 in 2027. We continue to believe investor debate will focus around 25-30x on $14+ in 2027. **We reiterate our Overweight rating & our $395 PT is based on 29x 2027E GAAP EPS of $13.60.**

- **AI Continues to Expand the Search Opportunity.** Google Search & Other accelerated to 17% growth in 4Q w/strength led by the retail, finance, and health verticals. Google shipped more than 250 search product launches within AI Overviews and AI Mode, and AI continues to drive higher overall search engagement. Search queries are becoming more complex, w/AI Mode queries 3x longer than traditional searches, and the search experience is becoming more conversational, w/a significant portion of queries resulting in follow-up questions. **Importantly, we believe AI is helping Google expand its ad coverage, and that is just getting started.** Google traditionally has shown ads against only ~20% of queries, but Gemini and AI Max are helping better match ads against more complex queries, thereby increasing coverage and the overall monetization opportunity. Search strength was offset slightly by softness in YouTube ads, which grew 9% Y/Y, and faced a tough comp from political in 4Q24, along w/some category softness.
- **Significant Cloud Acceleration.** Google Cloud had an excellent quarter with 48% growth well ahead of our +38% expectation, and exiting the quarter at a $70B+ run-rate. Enterprise AI products drove strength and are now generating billions of dollars in quarterly revenue. Nearly 75% of Google Cloud customers have used Google's vertically optimized AI, & products built on Google AI models increased nearly 400% Y/Y. Cloud is seeing very strong momentum across both new and existing customers, with new customer velocity exiting the year doubling from 1Q levels, and existing customers

Overweight

GOOG, GOOG US
Price (04 Feb 26):$333.34
▲**Price Target (Dec-26):$395.00**
Prior (Dec-26):$385.00

Overweight

GOOGL, GOOGL US
Price (04 Feb 26):$333.04
▲**Price Target (Dec-26):$395.00**
Prior (Dec-26):$385.00

Internet - Large Cap / Mid & Small Cap

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

Dae K Lee, CFA
(1-415) 315-8500
dae.k.lee@jpmorgan.com

Margaret Hoffman
(1-212) 622-1710
margaret.hoffman@jpmchase.com
J.P. Morgan Securities LLC

See page 13 for analyst certification and important disclosures.

J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

www.jpmorganmarkets.com

outpacing their initial commitments by more than 30%. Many did a double-take on the $240B in Cloud backlog, which grew 55% Q/Q & is bigger than Amazon's $200B backlog in 3Q25 (which should also increase meaningfully later today). Google noted enterprise AI products account for a meaningful part of the increase, and the higher commitments are coming from a diverse customer base.

- **Capex Doubling This Year.** Google capex is expected to double from $91B in 2025 to the $175B-$185B range in 2026. Google indicated capex composition would be similar to 2025, with ~60% toward machines and servers, and ~40% toward long duration assets such as data centers and network equipment. Also like 2025, more than half of Google's ML compute will go toward Cloud. **Still, Google indicated that it is supply constrained, and its biggest challenge is compute capacity.** In order to offset bigger investments and increased depreciation, Google emphasized that it has a mindset for ongoing efficiencies.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Summary of the Quarter

Figure 1: Alphabet, Inc. Actual vs. J.P. Morgan Estimates

$ in millions, except per share data and where noted otherwise	4Q25 JPMe	4Q25 Actuals	Diff (%/bps)
Google Services			
Google Search & other Revenue	61,780	63,073	2.1%
Y/Y Growth	14.3%	16.7%	239 bps
YouTube ads Revenue	11,922	11,383	-4.5%
Y/Y Growth	13.8%	8.7%	(514) bps
Google Network Revenue	7,772	7,828	0.7%
Y/Y Growth	-2.3%	-1.6%	70 bps
Google SPD Revenue	13,945	13,578	-2.6%
Google Services Revenue	95,419	95,862	0.5%
Y/Y Growth	13.5%	14.0%	53 bps
Operating Income / (Loss)	38,788	40,132	3.5%
Margin as % of Gross Revenue	40.7%	41.9%	121 bps
Google Cloud			
Revenue	16,499	17,664	7.1%
Y/Y Growth	38.0%	47.8%	974 bps
Operating Income / (Loss)	3,960	5,313	34.2%
% Margin - Gross	24.0%	30.1%	608 bps
Other Bets			
Revenue	364	370	1.6%
Operating Income / (Loss)	(1,580)	(3,617)	-128.9%
Reconciling Items			
Hedging gains (losses)	(164)	(68)	-58.5%
Corporate costs, unallocated	(3,869)	(5,894)	-52.3%
Alphabet Inc.			
Alphabet Gross Revenue	112,119	113,828	1.5%
FXHN Y/Y Growth	*15.6%*	*16.7%*	114 bps
Y/Y Growth	*16.2%*	*18.0%*	177 bps
Q/Q Growth	*9.5%*	*11.2%*	167 bps
Alphabet Net Revenue	95,884	97,231	1.4%
Y/Y Growth	*17.5%*	*19.1%*	165 bps
Q/Q Growth	*9.6%*	*11.2%*	154 bps
Alphabet GAAP Operating Income	37,299	35,934	-3.7%
as % of Gross Revenue	33.3%	31.6%	(170) bps
Alphabet GAAP EPS	$2.63	$2.82	7.0%
Y/Y Growth	*22.5%*	*31.1%*	861 bps

Source: J.P. Morgan estimates, Company data.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Adjusting Estimates

We are adjusting our estimates, as shown below.

Figure 2: Adjusting J.P. Morgan Estimates

Alphabet, Inc	4Q25 JPMe		1Q26 JPMe		2Q26 JPMe		3Q26 JPMe		4Q26 JPMe		2026 JPMe		2027 JPMe	
($ in millions, except where noted)	Old	New	Old	New	Old	New	Old	New	Old	New	Old	New	Old	New
Google Services														
Google Search & other Revenue	61,780	63,073	59,117	59,890	61,610	62,476	63,277	64,159	69,183	71,601	253,187	258,126	281,038	289,175
FXHN Y/Y Growth	***13.5%***	***15.4%***	***13.0%***	***14.0%***	***13.0%***	***14.0%***	***12.5%***	***13.5%***	***12.0%***	***13.0%***	***12.6%***	***13.6%***	***11.0%***	***12.0%***
Y/Y Growth	*14.3%*	*16.7%*	*16.6%*	*18.1%*	*13.7%*	*15.3%*	*11.9%*	*13.4%*	*12.0%*	*13.5%*	*13.4%*	*15.0%*	*11.0%*	*12.0%*
Q/Q Growth	*9.2%*	*11.5%*	*-4.3%*	*-5.0%*	*4.2%*	*4.3%*	*2.7%*	*2.7%*	*9.3%*	*11.6%*				
% chg vs. old		2.1%		1.3%		1.4%		1.4%		3.5%		2.0%		2.9%
YouTube ads Revenue	11,922	11,383	10,455	10,267	11,285	11,096	11,682	11,484	13,648	12,808	47,070	45,655	52,483	50,690
FXHN Y/Y Growth	***13.0%***	***7.4%***	***13.5%***	***11.0%***	***14.5%***	***12.0%***	***14.5%***	***12.0%***	***14.5%***	***12.0%***	***14.2%***	***11.8%***	***11.5%***	***11.0%***
Y/Y Growth	*13.8%*	*8.7%*	*17.1%*	*15.0%*	*15.2%*	*13.3%*	*13.9%*	*11.9%*	*14.5%*	*12.5%*	*15.1%*	*13.1%*	*11.5%*	*11.0%*
Q/Q Growth	*16.2%*	*10.9%*	*-12.3%*	*-9.8%*	*7.9%*	*8.1%*	*3.5%*	*3.5%*	*16.8%*	*11.5%*				
% chg vs. old		-4.5%		-1.8%		-1.7%		-1.7%		-6.2%		-3.0%		-3.4%
Google Network Revenue	7,772	7,828	7,337	7,368	7,251	7,288	7,166	7,202	7,616	7,707	29,370	29,565	29,076	28,981
Y/Y Growth	*-2.3%*	*-1.6%*	*1.1%*	*1.5%*	*-1.4%*	*-0.9%*	*-2.6%*	*-2.1%*	*-2.0%*	*-1.5%*	*-1.2%*	*-0.8%*	*-1.0%*	*-2.0%*
Q/Q Growth	*5.7%*	*6.4%*	*-5.6%*	*-5.9%*	*-1.2%*	*-1.1%*	*-1.2%*	*-1.2%*	*6.3%*	*7.0%*				
% chg vs. old		0.7%		0.4%		0.5%		0.5%		1.2%		0.7%		-0.3%
Google SPD Revenue	13,945	13,578	12,423	12,475	13,188	13,256	14,972	15,048	16,174	15,823	56,757	56,601	64,136	63,976
Y/Y Growth	*19.9%*	*16.7%*	*19.7%*	*20.2%*	*17.7%*	*18.3%*	*16.3%*	*16.9%*	*16.0%*	*16.5%*	*17.3%*	*17.8%*	*13.0%*	*13.0%*
Q/Q Growth	*8.4%*	*5.5%*	*-10.9%*	*-8.1%*	*6.2%*	*6.3%*	*13.5%*	*13.5%*	*8.0%*	*5.2%*				
% chg vs. old		-2.6%		0.4%		0.5%		0.5%		-2.2%		-0.3%		-0.2%
Google Services Revenue	95,419	95,862	89,332	90,000	93,334	94,116	97,097	97,893	106,621	107,939	386,384	389,948	426,733	432,822
FXHN Y/Y Growth	***12.6%***	***12.7%***	***12.1%***	***12.4%***	***12.4%***	***12.7%***	***12.2%***	***12.5%***	***11.8%***	***12.1%***	***12.1%***	***12.4%***	***10.4%***	***11.0%***
Y/Y Growth	*13.5%*	*14.0%*	*15.6%*	*16.5%*	*13.1%*	*14.0%*	*11.5%*	*12.5%*	*11.7%*	*12.6%*	*12.9%*	*13.8%*	*10.4%*	*11.0%*
Q/Q Growth	*9.6%*	*10.1%*	*-6.4%*	*-6.1%*	*4.5%*	*4.6%*	*4.0%*	*4.0%*	*9.8%*	*10.3%*				
% chg vs. old		0.5%		0.7%		0.8%		0.8%		1.2%		0.9%		1.4%
Total TAC as % of Ads Gross Revs	19.9%	20.2%	20.2%	20.2%	19.9%	19.9%	19.8%	19.8%	19.5%	19.7%	19.8%	19.9%	19.6%	19.6%
bps chg vs. old		24		4		5		5		19		9		6
Google Services GAAP OI	38,788	40,132	39,306	40,050	40,320	40,658	41,266	42,094	44,781	45,874	165,674	168,676	183,495	190,442
Margin as % of Gross Revenue	40.7%	41.9%	44.0%	44.5%	43.2%	43.2%	42.5%	43.0%	42.0%	42.5%	42.9%	43.3%	43.0%	44.0%
% chg vs. old		3.5%		1.9%		0.8%		2.0%		2.4%		1.8%		3.8%
Google Services GAAP EPS	$2.75	$2.97	$2.79	$2.84	$2.84	$2.84	$2.90	$2.92	$3.14	$3.16	$11.66	$11.76	$12.86	$13.03
Y/Y Growth	*21.1%*	*30.7%*	*27.5%*	*29.6%*	*20.4%*	*20.5%*	*32.3%*	*33.4%*	*13.8%*	*6.4%*	*22.9%*	*21.1%*	*10.3%*	*10.7%*
% chg vs. old		7.9%		1.6%		0.1%		0.8%		0.8%		0.8%		1.3%
Google Cloud														
Google Cloud Revenue	16,499	17,664	17,963	19,690	19,739	22,045	22,004	23,932	24,250	26,086	83,956	91,752	113,340	128,486
FXHN Y/Y Growth	***37.0%***	***46.0%***	***42.0%***	***55.0%***	***44.0%***	***60.0%***	***46.0%***	***58.0%***	***47.0%***	***47.0%***	***45.0%***	***54.5%***	***35.0%***	***40.0%***
Y/Y Growth	*38.0%*	*47.8%*	*46.5%*	*60.6%*	*44.9%*	*61.8%*	*45.2%*	*57.9%*	*47.0%*	*47.7%*	*45.9%*	*56.3%*	*35.0%*	*40.0%*
Q/Q Growth	*8.9%*	*16.5%*	*8.9%*	*11.5%*	*9.9%*	*12.0%*	*11.5%*	*8.6%*	*10.2%*	*9.0%*				
% chg vs. old		7.1%		9.6%		11.7%		8.8%		7.6%		9.3%		13.4%
Google Cloud GAAP OI	3,960	5,313	4,581	4,726	4,954	5,952	5,369	6,462	5,771	7,304	20,675	24,443	28,335	34,691
% Margin - Gross	24.0%	30.1%	25.5%	24.0%	25.1%	27.0%	24.4%	27.0%	23.8%	28.0%	24.6%	26.6%	25.0%	27.0%
% chg vs. old		34.2%		3.2%		20.1%		20.4%		26.6%		18.2%		22.4%
Google Cloud GAAP EPS	$0.32	$0.43	$0.37	$0.39	$0.41	$0.48	$0.44	$0.52	$0.47	$0.59	$1.69	$1.99	$2.32	$2.81
Y/Y Growth	*91.4%*	*156.3%*	*111.7%*	*117.7%*	*74.9%*	*109.2%*	*49.0%*	*78.2%*	*45.3%*	*36.3%*	*64.6%*	*74.8%*	*37.5%*	*41.6%*
% chg vs. old		34.0%		2.8%		19.6%		19.6%		25.6%		17.6%		21.1%
Other Bets														
Other Bets Revenue	364	370	425	470	454	514	477	505	497	545	1,852	2,035	2,073	3,080
% chg vs. old		1.6%		10.8%		13.3%		5.9%		9.7%		9.8%		48.6%
Other Bets GAAP OI	($1,580)	($3,617)	($1,239)	($1,406)	($1,254)	($1,443)	($1,384)	($1,478)	($1,500)	($1,797)	($5,377)	($6,124)	($5,458)	($5,662)
$ chg vs. old		($2,037)		($167)		($189)		($94)		($298)		($748)		($204)
Other Bets GAAP EPS	($0.13)	($0.11)	($0.10)	($0.11)	($0.10)	($0.12)	($0.11)	($0.12)	($0.12)	($0.15)	($0.44)	($0.50)	($0.45)	($0.46)
% chg vs. old		16.9%		-13.1%		-14.6%		-6.2%		-18.9%		-13.3%		-2.6%
Reconciling Items														
Hedging gains (losses)	(164)	(68)	(662)	(765)	(138)	(260)	136	17	4	(123)	(660)	(1,132)	0	(24)
$ chg vs. old		96		(103)		(122)		(119)		(127)		(472)		(24)
Corporate costs, unallocated	($3,869)	($5,894)	($4,971)	($5,172)	($4,509)	($4,752)	($4,528)	($4,754)	($4,725)	($4,968)	($18,732)	($19,645)	($21,035)	($21,922)
$ chg vs. old		($2,025)		($201)		($243)		($226)		($243)		($913)		($887)

Alphabet, Inc	4Q25 JPMe		1Q26 JPMe		2Q26 JPMe		3Q26 JPMe		4Q26 JPMe		2026 JPMe		2027 JPMe	
($ in millions, except where noted)	Old	New		New		New		New		New	Old	New	Old	New
Alphabet Inc.														
Alphabet Gross Revenue	112,119	113,828	107,058	109,394	113,389	116,415	119,713	122,346	131,372	134,447	471,532	482,603	542,146	564,365
FXHN Y/Y Growth	***15.6%***	***16.7%***	***16.0%***	***18.2%***	***16.9%***	***19.5%***	***17.3%***	***19.4%***	***17.0%***	***17.6%***	***16.8%***	***18.6%***	***14.8%***	***16.6%***
Y/Y Growth	*16.2%*	*18.0%*	*18.6%*	*21.2%*	*17.6%*	*20.7%*	*17.0%*	*19.5%*	*17.2%*	*18.1%*	*17.6%*	*19.8%*	*15.0%*	*16.9%*
Q/Q Growth	*9.5%*	*11.2%*	*-4.5%*	*-3.9%*	*5.9%*	*6.4%*	*5.6%*	*5.1%*	*9.7%*	*9.9%*				
% chg vs. old		1.5%		2.2%		2.7%		2.2%		2.3%		2.3%		4.1%
Alphabet Net Revenue	95,884	97,231	91,560	93,738	97,441	100,286	103,491	105,940	113,747	116,324	406,239	416,288	471,235	492,028
Y/Y Growth	*17.5%*	*19.1%*	*19.7%*	*22.6%*	*19.2%*	*22.7%*	*18.3%*	*21.1%*	*18.6%*	*19.6%*	*18.9%*	*21.4%*	*16.0%*	*18.2%*
Q/Q Growth	*9.6%*	*11.2%*	*-4.5%*	*-3.6%*	*6.4%*	*7.0%*	*6.2%*	*5.6%*	*9.9%*	*9.8%*				
% chg vs. old		1.4%		2.4%		2.9%		2.4%		2.3%		2.5%		4.4%
GAAP Operating Income	37,299	35,934	37,677	38,198	39,512	40,415	40,724	42,323	44,327	46,413	162,240	167,350	185,337	197,549
% Margin - Gross	33.3%	31.6%	35.2%	34.9%	34.8%	34.7%	34.0%	34.6%	33.7%	34.5%	34.4%	34.7%	34.2%	35.0%
% chg vs. old		-3.7%		1.4%		2.3%		3.9%		4.7%		3.1%		6.6%
GAAP EPS	$2.63	$2.82	$2.66	$2.69	$2.78	$2.82	$2.85	$2.94	$3.10	$3.21	$11.38	$11.66	$13.02	$13.60
Y/Y Growth	*22.5%*	*31.1%*	*-5.4%*	*-4.4%*	*20.1%*	*22.2%*	*-0.5%*	*2.5%*	*17.7%*	*13.8%*	*7.2%*	*7.8%*	*14.3%*	*16.7%*
% chg vs. old		7.0%		1.1%		1.7%		3.0%		3.4%		2.4%		4.5%
Capex	28,765	27,851	32,046	35,620	35,079	45,129	36,739	47,673	38,105	52,346	141,969	180,768	160,220	209,112
Y/Y Growth	*101.5%*	*95.1%*	*86.3%*	*107.1%*	*56.3%*	*101.1%*	*53.4%*	*99.0%*	*32.5%*	*87.9%*	*53.7%*	*97.7%*	*12.9%*	*15.7%*
% chg vs. old		-3.2%		11.2%		28.6%		29.8%		37.4%		27.3%		30.5%
FCF	19,052	24,551	18,446	17,977	2,925	(7,422)	21,544	11,920	22,113	6,308	65,028	28,782	83,086	44,492
Y/Y Growth	*-23.3%*	*-1.2%*	*-2.7%*	*-5.2%*	*-44.8%*	*-240.0%*	*-11.9%*	*-51.3%*	*16.1%*	*-74.3%*	*-4.0%*	*-60.7%*	*27.8%*	*54.6%*
% chg vs. old		28.9%		-2.5%		-353.7%		-44.7%		-71.5%		-55.7%		-46.5%

Source: J.P. Morgan estimates, Company data.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Model Pages

Figure 3: Alphabet Income Statement

$ in millions, except per share amounts	2024A	3/25A	6/25A	9/25A	12/25A	2025A	3/26E	6/26E	9/26E	12/26E	2026E	2027E
Gross Revenues	**350,018.0**	**90,234**	**96,428**	**102,346**	**113,828**	**402,836.0**	**109,394.5**	**116,415.2**	**122,346.1**	**134,447.2**	**482,603.0**	**564,364.8**
Traffic Acquisition Costs (TAC)	54,900.0	13,748	14,705	14,876	16,597	59,926.0	15,656.8	16,128.9	16,406.0	18,123.5	66,315.1	72,336.5
Net Revenues excluding TAC	**295,118.0**	**76,486**	**81,723**	**87,470**	**97,231**	**342,910.0**	**93,737.7**	**100,286.3**	**105,940.1**	**116,323.8**	**416,287.9**	**492,028.3**
Other cost of revenues	91,406.0	22,613	24,334	26,493	29,169	102,609.0	28,074.5	30,386.7	32,735.5	35,827.7	127,024.4	156,957.0
Gross profit	**203,712.0**	**53,873**	**57,389**	**60,977**	**68,062**	**240,301.0**	**65,663.3**	**69,899.5**	**73,204.6**	**80,496.0**	**289,263.5**	**335,071.2**
Operating expenses			R&D ex. Waymo comp charge		16,472							
Research & development	49,326.0	13,556	13,808	15,151	18,572	61,087.0	16,872.8	17,149.0	18,327.6	19,077.1	71,426.5	81,676.7
Sales & marketing	27,808.0	6,172	7,101	7,205	8,215	28,693.0	6,936.6	7,922.6	8,051.4	9,073.3	31,983.9	35,180.0
General & administrative	14,188.0	3,539	5,209	3,936	5,341	18,025.0	3,655.8	4,412.6	4,502.5	5,932.5	18,503.3	20,665.2
Other 1x item, incl. Impairment charge	0.0	0	0	3,457	0	3,457.0	0.0	0.0	0.0	0.0	0.0	
Total operating expenses	91,322.0	23,267	26,118	29,749	32,128	111,262.0	27,465.2	29,484.2	30,881.5	34,082.9	121,913.7	137,521.9
GAAP Operating income	**112,390.0**	**30,606**	**31,271**	**31,228**	**35,934**	**129,039.0**	**38,198.1**	**40,415.4**	**42,323.1**	**46,413.2**	**167,349.8**	**197,549.3**
Non-GAAP Operating income	**112,390.0**	**30,606**	**31,271**	**31,228**	**35,934**	**129,039.0**	**38,198.1**	**40,415.4**	**42,323.1**	**46,413.2**	**167,349.8**	**197,549.3**
JPMe Adjusted Operating income	**135,175.0**	**36,122**	**37,269**	**41,053**	**43,005**	**157,449.0**	**44,478.6**	**47,335.1**	**49,738.9**	**53,974.2**	**195,526.8**	**229,039.2**
Other income, expense and other, net	7,425.0	11,183	2,662	12,759	3,183	29,787.0	979.6	872.6	733.0	667.0	3,252.1	2,075.5
Income before taxes	119,815.0	41,789	33,933	43,987	39,117	158,826.0	39,177.7	41,287.9	43,056.0	47,080.2	170,601.9	199,624.9
Provision for income taxes	19,697.0	7,249	5,737	9,008	4,662	26,656.0	6,268.4	6,606.1	6,889.0	7,532.8	27,296.3	31,940.0
Effective tax rate	*16.4%*	*17.3%*	*16.9%*	*20.5%*	*11.9%*	*16.8%*	*16.0%*	*16.0%*	*16.0%*	*16.0%*	*16.0%*	*16.0%*
Discontinued Operations - Motorola												
GAAP Net Income (loss)	**100,118.0**	**34,540**	**28,196**	**34,979**	**34,455**	**132,170.0**	**32,909.3**	**34,681.9**	**36,167.1**	**39,547.4**	**143,305.6**	**167,684.9**
GAAP EPS - Diluted	**$8.04**	**$2.81**	**$2.31**	**$2.87**	**$2.82**	**$10.81**	**$2.69**	**$2.82**	**$2.94**	**$3.21**	**$11.66**	**$13.60**
Diluted shares outstanding	12,447.3	12,291	12,198	12,203	12,228	12,230.0	12,254.9	12,281.8	12,308.9	12,336.1	12,295.4	12,326.8
Alphabet Non-GAAP reconciliation												
Adj. for One-time item	0.0	0.0	0.0	3,457.0	0.0	3,457.0	0.0	0.0	0.0	0.0	0.0	0.0
Non-GAAP Net Income (loss)	**100,118.0**	**34,540.0**	**28,196.0**	**38,436.0**	**34,455.0**	**135,627.0**	**32,909.3**	**34,681.9**	**36,167.1**	**39,547.4**	**143,305.6**	**167,684.9**
Non-GAAP EPS - Diluted	**$8.04**	**$2.81**	**$2.31**	**$3.15**	**$2.82**	**$11.09**	**$2.69**	**$2.82**	**$2.94**	**$3.21**	**$11.66**	**$13.60**
Non-recurring items:												
Stock compensation expense	22,785.0	5,516.0	5,998.0	6,368.0	7,071.0	24,953.0	6,280.4	6,919.8	7,415.8	7,561.0	28,177.0	31,489.8
One-time items	(2,943.0)	(10,182.0)	(1,612.0)	(8,226.0)	(1,973.0)	(21,993.0)	397.3	397.3	397.3	397.3	1,589.3	1,545.7
Tax Effect	(3,208.3)	809.4	(741.5)	380.5	(607.6)	(159.2)	(1,068.4)	(1,170.7)	(1,250.1)	(1,273.3)	(4,762.6)	(5,285.7)
Adjusted Net Income (loss)	**116,751.7**	**30,683.4**	**31,840.5**	**33,501.5**	**38,945.4**	**134,970.8**	**38,518.6**	**40,828.2**	**42,730.1**	**46,232.4**	**168,309.3**	**195,434.7**
Adj. EPS - Diluted	**$9.38**	**$2.50**	**$2.61**	**$2.75**	**$3.18**	**$11.04**	**$3.14**	**$3.32**	**$3.47**	**$3.75**	**$13.69**	**$15.85**
Diluted shares outstanding	12,447.3	12,291.0	12,198.0	12,203.0	12,228.0	12,230.0	12,254.9	12,281.8	12,308.9	12,336.1	12,295.4	12,326.8
Adj. EBITDA Reconciliation												
Operating income (adj. for 1x item as noted)	112,390.0	30,606	31,271	34,685	35,934	132,496.0	38,198.1	40,415.4	42,323.1	46,413.2	167,349.8	197,549.3
Depreciation & amortization	15,311.0	4,487	4,998	5,611	6,040	21,136.0	7,030.3	8,223.5	9,746.5	11,632.4	36,632.7	53,631.1
Stock compensation expense	22,785.0	5,516	5,998	6,368	7,071	24,953.0	6,280.4	6,919.8	7,415.8	7,561.0	28,177.0	31,489.8
Adj. EBITDA	**150,486.0**	**40,609**	**42,267**	**46,664**	**49,045**	**178,585.0**	**51,508.9**	**55,558.6**	**59,485.4**	**65,606.6**	**232,159.5**	**282,670.2**
Year/Year Growth	**2024A**	**3/25A**	**6/25A**	**9/25A**	**12/25A**	**2025A**	**3/26E**	**6/26E**	**9/26E**	**12/26E**	**2026E**	**2027E**
Gross Revenues (FXHN)	***14.8%***	***13.8%***	***13.4%***	***14.7%***	***16.7%***	***14.7%***	***18.2%***	***19.5%***	***19.4%***	***17.6%***	***18.6%***	***16.6%***
Gross Revenues	***13.9%***	***12.0%***	***13.8%***	***15.9%***	***18.0%***	***15.1%***	***21.2%***	***20.7%***	***19.5%***	***18.1%***	***19.8%***	***16.9%***
Traffic Acquisition Costs (TAC)	*7.9%*	*6.2%*	*9.8%*	*8.4%*	*11.8%*	*9.2%*	*13.9%*	*9.7%*	*10.3%*	*9.2%*	*10.7%*	*9.1%*
Net Revenues	***15.1%***	***13.2%***	***14.5%***	***17.3%***	***19.1%***	***16.2%***	***22.6%***	***22.7%***	***21.1%***	***19.6%***	***21.4%***	***18.2%***
Other cost of revenues	*10.9%*	*8.9%*	*10.0%*	*16.4%*	*13.2%*	*12.3%*	*24.2%*	*24.9%*	*23.6%*	*22.8%*	*23.8%*	*23.6%*
Gross profit	***17.0%***	***15.0%***	***16.6%***	***17.7%***	***21.9%***	***18.0%***	***21.9%***	***21.8%***	***20.1%***	***18.3%***	***20.4%***	***15.8%***
Research & development	*8.6%*	*13.9%*	*16.4%*	*21.7%*	*41.6%*	*23.8%*	*24.5%*	*24.2%*	*21.0%*	*2.7%*	*16.9%*	*14.4%*
Sales & marketing	*-0.4%*	*-4.0%*	*4.5%*	*-0.3%*	*11.6%*	*3.2%*	*12.4%*	*11.6%*	*11.7%*	*10.4%*	*11.5%*	*10.0%*
General & administrative	*-13.6%*	*17.0%*	*64.9%*	*9.4%*	*21.2%*	*27.0%*	*3.3%*	*-15.3%*	*14.4%*	*11.1%*	*2.7%*	*11.7%*
Total operating expenses	***1.7%***	***9.0%***	***19.8%***	***27.8%***	***29.1%***	***21.8%***	***18.0%***	***12.9%***	***3.8%***	***6.1%***	***9.6%***	***12.8%***
GAAP Operating income	***33.3%***	***20.2%***	***14.0%***	***9.5%***	***16.0%***	***14.8%***	***24.8%***	***29.2%***	***35.5%***	***29.2%***	***29.7%***	***18.0%***
Non-GAAP Operating income	***33.3%***	***20.2%***	***14.0%***	***9.5%***	***16.0%***	***14.8%***	***24.8%***	***29.2%***	***35.5%***	***29.2%***	***29.7%***	***18.0%***
JPMe Adjusted Operating income	***26.6%***	***17.5%***	***12.0%***	***19.5%***	***16.9%***	***16.5%***	***23.1%***	***27.0%***	***21.2%***	***25.5%***	***24.2%***	***17.1%***
Depreciation & amortization	*28.2%*	*31.5%*	*34.8%*	*40.8%*	*43.6%*	*38.0%*	*56.7%*	*64.5%*	*73.7%*	*92.6%*	*73.3%*	*46.4%*
Stock compensation expense	*1.4%*	*4.8%*	*2.3%*	*8.9%*	*21.7%*	*9.5%*	*13.9%*	*15.4%*	*16.5%*	*6.9%*	*12.9%*	*11.8%*
EBITDA	*24.9%*	*18.9%*	*14.2%*	*21.7%*	*19.7%*	*18.7%*	*26.8%*	*31.4%*	*27.5%*	*33.8%*	*30.0%*	*21.8%*
GAAP Net Income (loss)	*35.7%*	*46.0%*	*19.4%*	*33.0%*	*29.8%*	*32.0%*	*-4.7%*	*23.0%*	*3.4%*	*14.8%*	*8.4%*	*17.0%*
GAAP EPS	***38.7%***	***48.8%***	***22.3%***	***35.3%***	***31.1%***	***34.4%***	***-4.4%***	***22.2%***	***2.5%***	***13.8%***	***7.8%***	***16.7%***
Non-GAAP Net Income (loss)	*35.7%*	*46.0%*	*19.4%*	*46.1%*	*29.8%*	*35.5%*	*-4.7%*	*23.0%*	*-5.9%*	*14.8%*	*5.7%*	*17.0%*
Non-GAAP EPS	*38.7%*	*48.8%*	*22.3%*	*48.7%*	*31.1%*	*37.9%*	*-4.4%*	*22.2%*	*-6.7%*	*13.8%*	*5.1%*	*16.7%*
JPMe Adjusted Net Income (loss)	*22.5%*	*15.5%*	*8.0%*	*13.4%*	*25.0%*	*15.6%*	*25.5%*	*28.2%*	*27.5%*	*18.7%*	*24.7%*	*16.1%*
JPMe Adjusted EPS	*25.2%*	*17.7%*	*10.7%*	*15.4%*	*26.2%*	*17.7%*	*25.9%*	*27.4%*	*26.4%*	*17.7%*	*24.0%*	*15.8%*
GAAP expenses as % of Net Revenues												
Other cost of revenues (non-TAC)	31.0%	29.6%	29.8%	30.3%	30.0%	29.9%	30.0%	30.3%	30.9%	30.8%	30.5%	31.9%
Research & development	16.7%	17.7%	16.9%	17.3%	19.1%	17.8%	18.0%	17.1%	17.3%	16.4%	17.2%	16.6%
Sales & marketing	9.4%	8.1%	8.7%	8.2%	8.4%	8.4%	7.4%	7.9%	7.6%	7.8%	7.7%	7.2%
General & administrative	4.8%	4.6%	6.4%	4.5%	5.5%	5.3%	3.9%	4.4%	4.3%	5.1%	4.4%	4.2%
Stock compensation expense	7.7%	7.2%	7.3%	7.3%	7.3%	7.3%	6.7%	6.9%	7.0%	6.5%	6.8%	6.4%
GAAP expenses as % of Gross Revenues												
Traffic Acquisition Costs (TAC)	15.7%	15.2%	15.2%	14.5%	14.6%	14.9%	14.3%	13.9%	13.4%	13.5%	13.7%	12.8%
Other cost of revenues (non-TAC)	26.1%	25.1%	25.2%	25.9%	25.6%	25.5%	25.7%	26.1%	26.8%	26.6%	26.3%	27.8%
Research & development	14.1%	15.0%	14.3%	14.8%	16.3%	15.2%	15.4%	14.7%	15.0%	14.2%	14.8%	14.5%
Sales & marketing	7.9%	6.8%	7.4%	7.0%	7.2%	7.1%	6.3%	6.8%	6.6%	6.7%	6.6%	6.2%
General & administrative	4.1%	3.9%	5.4%	3.8%	4.7%	4.5%	3.3%	3.8%	3.7%	4.4%	3.8%	3.7%
Stock compensation expense	6.5%	6.1%	6.2%	6.2%	6.2%	6.2%	5.7%	5.9%	6.1%	5.6%	5.8%	5.6%
Margin as % of Gross Revenues												
Gross margin	58.2%	59.7%	59.5%	59.6%	59.8%	59.7%	60.0%	60.0%	59.8%	59.9%	59.9%	59.4%
GAAP Operating margin	**32.1%**	**33.9%**	**32.4%**	**30.5%**	**31.6%**	**32.0%**	**34.9%**	**34.7%**	**34.6%**	**34.5%**	**34.7%**	**35.0%**
Non-GAAP Operating margin	**32.1%**	**33.9%**	**32.4%**	**30.5%**	**31.6%**	**32.0%**	**34.9%**	**34.7%**	**34.6%**	**34.5%**	**34.7%**	**35.0%**
JPMe Adjusted Operating margin	**38.6%**	**40.0%**	**38.6%**	**40.1%**	**37.8%**	**39.1%**	**40.7%**	**40.7%**	**40.7%**	**40.1%**	**40.5%**	**40.6%**
Net margin	28.6%	38.3%	29.2%	34.2%	30.3%	32.8%	30.1%	29.8%	29.6%	29.4%	29.7%	29.7%
Adj. EBITDA	**43.0%**	**45.0%**	**43.8%**	**45.6%**	**43.1%**	**44.3%**	**47.1%**	**47.7%**	**48.6%**	**48.8%**	**48.1%**	**50.1%**
Operating Leverage/Flow-through	**70.5%**	**66.6%**	**45.1%**	**59.0%**	**46.4%**	**53.2%**	**56.9%**	**66.5%**	**64.1%**	**80.3%**	**67.2%**	**61.8%**

Source: J.P. Morgan estimates, Company data.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Figure 4: Alphabet Revenue Drivers

$ millions except where noted	2024A	3/25A	6/25A	9/25A	12/25A	2025A	3/26E	6/26E	9/26E	12/26E	2026E	2027E
Gross Revenue Summary												
Google Search & other	198,084.0	50,702	54,190	56,567	63,073	224,532.0	59,889.6	62,475.7	64,159.2	71,601.5	258,126.0	289,175.0
Y/Y growth	*13.2%*	*9.8%*	*11.7%*	*14.5%*	*16.7%*	*13.4%*	*18.1%*	*15.3%*	*13.4%*	*13.5%*	*15.0%*	*12.0%*
FXHN Y/Y growth (estimate)	*14.1%*	*11.8%*	*11.1%*	*13.0%*	*15.4%*	*12.9%*	*14.0%*	*14.0%*	*13.5%*	*13.0%*	*13.6%*	*12.0%*
YouTube ads	36,147.0	8,927	9,796	10,261	11,383	40,367.0	10,267.2	11,095.7	11,484.4	12,807.8	45,655.0	50,690.0
Y/Y growth	*14.7%*	*10.3%*	*13.1%*	*15.0%*	*8.7%*	*11.7%*	*15.0%*	*13.3%*	*11.9%*	*12.5%*	*13.1%*	*11.0%*
FXHN Y/Y growth (estimate)	*15.7%*	*12.3%*	*12.4%*	*13.5%*	*7.4%*	*11.2%*	*11.0%*	*12.0%*	*12.0%*	*12.0%*	*11.8%*	*11.0%*
Google Properties	234,231.0	59,629	63,986	66,828	74,456	264,899.0	70,156.8	73,571.4	75,643.6	84,409.3	303,781.0	339,865.1
Y/Y growth	*13.4%*	*9.9%*	*11.9%*	*14.6%*	*15.4%*	*13.1%*	*17.7%*	*15.0%*	*13.2%*	*13.4%*	*14.7%*	*11.9%*
FXHN Y/Y growth (estimate)	*14.4%*	*11.9%*	*11.3%*	*13.1%*	*14.1%*	*12.6%*	*13.6%*	*13.7%*	*13.3%*	*12.8%*	*13.3%*	*11.8%*
Q/Q growth		*-7.6%*	*7.3%*	*4.4%*	*11.4%*		*-5.8%*	*4.9%*	*2.8%*	*11.6%*		
Google Network Members' properties	30,359.0	7,256	7,354	7,354	7,828	29,792.0	7,367.9	7,288.5	7,201.9	7,706.8	29,565.2	28,981.3
Y/Y growth	*-3.0%*	*-2.1%*	*-1.2%*	*-2.6%*	*-1.6%*	*-1.9%*	*1.5%*	*-0.9%*	*-2.1%*	*-1.5%*	*-0.8%*	*-2.0%*
FXHN Y/Y growth (estimate)	*-2.2%*	*-0.4%*	*-1.8%*	*-3.8%*	*-2.7%*	*-2.2%*	*-2.0%*	*-2.0%*	*-2.0%*	*-2.0%*	*-2.0%*	*-2.0%*
Q/Q growth		*-8.8%*	*1.4%*	*0.0%*	*6.4%*		*-5.9%*	*-1.1%*	*-1.2%*	*7.0%*		
Google advertising	264,590.0	66,885	71,340	74,182	82,284	294,691.0	77,524.7	80,859.8	82,845.5	92,116.1	333,346.2	368,846.4
Y/Y growth	*11.2%*	*8.5%*	*10.4%*	*12.6%*	*13.6%*	*11.4%*	*15.9%*	*13.3%*	*11.7%*	*11.9%*	*13.1%*	*10.6%*
FXHN Y/Y growth	*12.2%*	*10.4%*	*9.8%*	*11.2%*	*12.2%*	*10.9%*	*11.9%*	*12.1%*	*11.8%*	*11.4%*	*11.8%*	*10.6%*
Google subscriptions, platforms, and devices	40,340.0	10,379	11,203	12,870	13,578	48,030.0	12,474.8	13,255.8	15,047.5	15,823.2	56,601.4	63,975.9
Y/Y growth	*16.3%*	*18.8%*	*20.3%*	*20.8%*	*16.7%*	*19.1%*	*20.2%*	*18.3%*	*16.9%*	*16.5%*	*17.8%*	*13.0%*
FXHN Y/Y growth (estimate)	*17.3%*	*20.9%*	*19.6%*	*19.2%*	*15.4%*	*18.6%*	*16.0%*	*17.0%*	*17.0%*	*16.0%*	*16.5%*	*13.0%*
Q/Q growth		*-10.8%*	*7.9%*	*14.9%*	*5.5%*		*-8.1%*	*6.3%*	*13.5%*	*5.2%*		
Google Services total	304,930.0	77,264	82,543	87,052	95,862	342,721.0	89,999.6	94,115.7	97,893.1	107,939.3	389,947.6	432,822.3
Y/Y growth	*11.9%*	*9.8%*	*11.7%*	*13.8%*	*14.0%*	*12.4%*	*16.5%*	*14.0%*	*12.5%*	*12.6%*	*13.8%*	*11.0%*
FXHN Y/Y growth	*12.8%*	*11.7%*	*11.0%*	*12.3%*	*12.7%*	*12.0%*	*12.4%*	*12.7%*	*12.5%*	*12.1%*	*12.4%*	*11.0%*
Google Cloud	43,229.0	12,260	13,624	15,157	17,664	58,705.0	19,689.9	22,045.1	23,931.5	26,085.9	91,752.4	128,486.3
Y/Y growth	*30.6%*	*28.1%*	*31.7%*	*33.5%*	*47.8%*	*35.8%*	*60.6%*	*61.8%*	*57.9%*	*47.7%*	*56.3%*	*40.0%*
FXHN Y/Y growth (estimate)	*31.8%*	*30.4%*	*30.9%*	*31.8%*	*46.0%*	*35.2%*	*55.0%*	*60.0%*	*58.0%*	*47.0%*	*54.5%*	*40.0%*
Q/Q growth		*2.6%*	*11.1%*	*11.3%*	*16.5%*		*11.5%*	*12.0%*	*8.6%*	*9.0%*		
Other Bets	1,648.0	450	373	344	370	1,537.0	470.4	514.4	504.7	545.1	2,034.6	3,080.1
Y/Y growth	*7.9%*	*-9.1%*	*2.2%*	*-11.3%*	*-7.5%*	*-6.7%*	*4.5%*	*37.9%*	*46.7%*	*47.3%*	*32.4%*	*51.4%*
Hedging gains (losses)	211.0	260	(112)	(207)	(68)	(127.0)	(765.3)	(260.0)	16.8	(123.2)	(1,131.6)	(23.9)
Alphabet total revenues (gross)	350,018.0	90,234	96,428	102,346	113,828	402,836.0	109,394.5	116,415.2	122,346.1	134,447.2	482,603.0	564,364.8
Y/Y growth	***13.9%***	***12.0%***	***13.8%***	***15.9%***	***18.0%***	***15.1%***	***21.2%***	***20.7%***	***19.5%***	***18.1%***	***19.8%***	***16.9%***
FXHN Y/Y growth	***14.8%***	***13.8%***	***13.4%***	***14.7%***	***16.7%***	***14.7%***	***18.2%***	***19.5%***	***19.4%***	***17.6%***	***18.6%***	***16.6%***
Q/Q growth		*-6.5%*	*6.9%*	*6.1%*	*11.2%*		*-3.9%*	*6.4%*	*5.1%*	*9.9%*		
Seasonality		22.4%	23.9%	25.4%	28.3%		22.7%	24.1%	25.4%	27.9%		
Gross Revenue Share by Segment												
Google Search & other	56.6%	56.2%	56.2%	55.3%	55.4%	55.7%	54.7%	53.7%	52.4%	53.3%	53.5%	51.2%
YouTube ads	10.3%	9.9%	10.2%	10.0%	10.0%	10.0%	9.4%	9.5%	9.4%	9.5%	9.5%	9.0%
Google Network Members' properties	8.7%	8.0%	7.6%	7.2%	6.9%	7.4%	6.7%	6.3%	5.9%	5.7%	6.1%	5.1%
Google advertising	**75.6%**	**74.1%**	**74.0%**	**72.5%**	**72.3%**	**73.2%**	**70.9%**	**69.5%**	**67.7%**	**68.5%**	**69.1%**	**65.4%**
Google subscriptions, platforms, and devices	11.5%	11.5%	11.6%	12.6%	11.9%	11.9%	11.4%	11.4%	12.3%	11.8%	11.7%	11.3%
Google Services total	**87.1%**	**85.6%**	**85.6%**	**85.1%**	**84.2%**	**85.1%**	**82.3%**	**80.8%**	**80.0%**	**80.3%**	**80.8%**	**76.7%**
Google Cloud	**12.4%**	**13.6%**	**14.1%**	**14.8%**	**15.5%**	**14.6%**	**18.0%**	**18.9%**	**19.6%**	**19.4%**	**19.0%**	**22.8%**
Other Bets	0.5%	0.5%	0.4%	0.3%	0.3%	0.4%	0.4%	0.4%	0.4%	0.4%	0.4%	0.5%
Hedging gains (losses)	0.1%	0.3%	-0.1%	-0.2%	-0.1%	0.0%	-0.7%	-0.2%	0.0%	-0.1%	-0.2%	0.0%
TAC Analysis												
Total Traffic Acquisition Costs (TAC)	**54,900.0**	**13,748**	**14,705**	**14,876**	**16,597**	**59,926.0**	**15,656.8**	**16,128.9**	**16,406.0**	**18,123.5**	**66,315.1**	**72,336.5**
Y/Y growth	*7.9%*	*6.2%*	*9.8%*	*8.4%*	*11.8%*	*9.2%*	*13.9%*	*9.7%*	*10.3%*	*9.2%*	*10.7%*	*9.1%*
TAC as a % of Total Gross Adv. Revenues	**20.7%**	**20.6%**	**20.6%**	**20.1%**	**20.2%**	**20.3%**	**20.2%**	**19.9%**	**19.8%**	**19.7%**	**19.9%**	**19.6%**
TAC as a % of Total Ads Gross Revs ex. YT	**24.0%**	**23.7%**	**23.9%**	**23.3%**	**23.4%**	**23.6%**	**23.3%**	**23.1%**	**23.0%**	**22.9%**	**23.1%**	**22.7%**
Net Revenue Summary												
Google Properties - Net Revenues	200,159.4	50,756.2	54,394.4	56,815.6	63,197.5	225,163.6	59,436.5	62,325.7	64,062.9	71,449.4	257,274.5	286,946.0
Y/Y growth	*13.0%*	*9.6%*	*11.1%*	*14.2%*	*14.7%*	*12.5%*	*17.1%*	*14.6%*	*12.8%*	*13.1%*	*14.3%*	*11.5%*
Q/Q growth		*-7.9%*	*7.2%*	*4.5%*	*11.2%*		*-6.0%*	*4.9%*	*2.8%*	*11.5%*		
Google Network Members' Properties - Net Revenu	9,530.6	2,380.9	2,240.6	2,490.4	2,489.5	9,601.4	2,431.4	2,405.2	2,376.6	2,543.3	9,756.5	9,563.8
Y/Y growth	*-2.9%*	*-1.0%*	*-0.6%*	*5.0%*	*-0.3%*	*0.7%*	*2.1%*	*7.3%*	*-4.6%*	*2.2%*	*1.6%*	*-2.0%*
Google Advertising - Net Revenues	**209,690.0**	**53,137.0**	**56,635.0**	**59,306.0**	**65,687.0**	**234,765.0**	**61,868.0**	**64,730.9**	**66,439.5**	**73,992.7**	**267,031.1**	**296,509.9**
Y/Y growth	*12.2%*	*9.1%*	*10.6%*	*13.8%*	*14.0%*	*12.0%*	*16.4%*	*14.3%*	*12.0%*	*12.6%*	*13.7%*	*11.0%*
Non-Advertising - Net Revenues	**85,428.0**	**23,349.0**	**25,088.0**	**28,164.0**	**31,544.0**	**108,145.0**	**31,869.8**	**35,555.4**	**39,500.6**	**42,331.1**	**149,256.8**	**195,518.4**
Y/Y growth	*22.8%*	*23.7%*	*24.7%*	*25.7%*	*31.4%*	*26.6%*	*36.5%*	*41.7%*	*40.3%*	*34.2%*	*38.0%*	*31.0%*
Alphabet Net Revenues	**295,118.0**	**76,486.0**	**81,723.0**	**87,470.0**	**97,231.0**	**342,910.0**	**93,737.7**	**100,286.3**	**105,940.1**	**116,323.8**	**416,287.9**	**492,028.3**
Y/Y growth	*15.1%*	*13.2%*	*14.5%*	*17.3%*	*19.1%*	*16.2%*	*22.6%*	*22.7%*	*21.1%*	*19.6%*	*21.4%*	*18.2%*
Q/Q growth		*-6.3%*	*6.8%*	*7.0%*	*11.2%*		*-3.6%*	*7.0%*	*5.6%*	*9.8%*		
Seasonality		22.3%	23.8%	25.5%	28.4%		22.5%	24.1%	25.4%	27.9%		
Net Revenue Share by Segment												
Google Properties as % of Alphabet Net Revenues	67.8%	66.4%	66.6%	65.0%	65.0%	65.7%	63.4%	62.1%	60.5%	61.4%	61.8%	58.3%
Google Network Members' Properties as % of Alphab	3.2%	3.1%	2.7%	2.8%	2.6%	2.8%	2.6%	2.4%	2.2%	2.2%	2.3%	1.9%
Google Advertising as % of Alphebt Net Revenues	**71.1%**	**69.5%**	**69.3%**	**67.8%**	**67.6%**	**68.5%**	**66.0%**	**64.5%**	**62.7%**	**63.6%**	**64.1%**	**60.3%**
Google Other & Other Bets as % of Alphabet Net F	**28.9%**	**30.5%**	**30.7%**	**32.2%**	**32.4%**	**31.5%**	**34.0%**	**35.5%**	**37.3%**	**36.4%**	**35.9%**	**39.7%**

($ millions except where noted)	2024A	3/25A	6/25A	9/25A	12/25A	2025A	3/26E	6/26E	9/26E	12/26E	2026E	2027E
MONETIZATION												
Google Search & Other												
Paid Clicks Change												
Y/Y growth	5%	2%	4%	7%								
QTD Y/Y growth		2%	3%	4%								
Cost-per-Click Change												
Y/Y growth	7%	7%	8%	7%								
QTD Y/Y growth		7%	8%	7%								
Google Network												
Impressions Change												
Y/Y growth	-11%	-5%	-6%	-5%								
Cost-per-Impression Change												
Y/Y growth	10%	4%	6%	6%								

Source: J.P. Morgan estimates, Company data.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Figure 5: Alphabet Revenue Drivers (Cont'd)

($ millions except where noted)	2024A	3/25A	6/25A	9/25A	12/25A	2025A	3/26E	6/26E	9/26E	12/26E	2026E	2027E
Geographic Breakdown												
Alphabet total revenues (gross)	350,018.0	90,234	96,428	102,346	113,828	402,836.0	109,394.5	116,415.2	122,346.1	134,447.2	482,603.0	564,364.8
Y/Y growth	*13.9%*	*12%*	*14%*	*16%*	*18%*	*15.1%*	*21.2%*	*20.7%*	*19.5%*	*18.1%*	*19.8%*	*16.9%*
Y/Y growth (ex-FX & Hedging)	***14.8%***	***14%***	***13%***	***15%***	***17%***	***14.7%***	***18.2%***	***19.5%***	***19.4%***	***17.6%***	***18.6%***	***16.6%***
Q/Q growth		*-6%*	*7%*	*6%*	*11%*		*-3.9%*	*6.4%*	*5.1%*	*9.9%*		
Geographic Breakdown												
United States	170,447.0	43,964	46,063	48,758	55,444	194,229.0	51,437.9	53,985.8	57,144.4	64,037.8	226,605.9	261,050.0
Y/Y growth	***16.5%***	***13%***	***12%***	***13%***	***17%***	***14.0%***	***17.0%***	***17.2%***	***17.2%***	***15.5%***	***16.7%***	***15.2%***
Q/Q growth		*-7%*	*5%*	*6%*	*14%*		*-7.2%*	*5.0%*	*5.9%*	*12.1%*		
U.S. revenues as % of Gross Revenues	48.7%	49%	48%	48%	49%	48.2%	47.0%	46.4%	46.7%	47.6%	47.0%	46.3%
International	179,360.0	46,010	50,477	53,795	58,452	208,734.0	58,721.9	62,689.3	65,184.9	70,532.6	257,128.7	303,338.7
Y/Y growth	*11.5%*	*10%*	*16%*	*19%*	*19%*	*16.4%*	*27.6%*	*24.2%*	*21.2%*	*20.7%*	*23.2%*	*18.0%*
FXHN Y/Y growth	*13.3%*	*14%*	*15%*	*16%*	*16%*	*15.5%*	*19.3%*	*21.6%*	*21.3%*	*19.6%*	*20.5%*	*17.9%*
Q/Q growth		*-6%*	*10%*	*7%*	*9%*		*0.5%*	*6.8%*	*4.0%*	*8.2%*		
International revenues as % of Gross Revenues	51.2%	51%	52%	53%	51%	51.8%	53.7%	53.8%	53.3%	52.5%	53.3%	53.7%
Hedging Gains (Losses)	211.0	260.0	(112.0)	(207.0)	(68.0)	(127.0)	(765.3)	(260.0)	16.8	(123.2)	(1,131.6)	(23.9)
Internation Breakdown (excl. Hedge post-1Q17 change)												
EMEA	102,127.0	25,923	28,262	29,911	33,056	117,152.0	32,918.0	34,262.0	34,969.7	38,436.2	140,585.8	161,003.1
Y/Y growth	*12.2%*	*9%*	*14%*	*17%*	*17%*	*14.7%*	*27.0%*	*21.2%*	*16.9%*	*16.3%*	*20.0%*	*14.5%*
FXHN Y/Y growth	***12.1%***	***12%***	***11%***	***12%***	***12%***	***12.0%***	***14.5%***	***16.4%***	***15.8%***	***14.5%***	***15.3%***	***14.5%***
Q/Q growth		*-8%*	*9%*	*6%*	*11%*		*-0.4%*	*4.1%*	*2.1%*	*9.9%*		
EMEA as a % of Gross Revenues	29.2%	29%	29%	29%	29%	29.1%	30.1%	29.4%	28.6%	28.6%	29.1%	28.5%
APAC	56,815.0	14,854	16,480	17,819	18,527	67,680.0	18,910.2	20,974.6	22,453.9	23,419.8	85,758.5	103,854.9
Y/Y growth	*10.3%*	*12%*	*19%*	*22%*	*22%*	*19.1%*	*27.3%*	*27.3%*	*26.0%*	*26.4%*	*26.7%*	*21.1%*
FXHN Y/Y growth	***12.9%***	***15%***	***18%***	***22%***	***23%***	***19.7%***	***25.0%***	***28.0%***	***28.0%***	***26.0%***	***26.8%***	***21.0%***
Q/Q growth		*-2%*	*11%*	*8%*	*4%*		*2.1%*	*10.9%*	*7.1%*	*4.3%*		
APAC as a % of Gross Revenues	16.2%	16%	17%	17%	16%	16.8%	17.3%	18.0%	18.4%	17.4%	17.8%	18.4%
Other Americas (Canada and LatAM)	20,418.0	5,233	5,735	6,065	6,869	23,902.0	6,893.8	7,452.7	7,761.4	8,676.6	30,784.5	38,480.6
Y/Y growth	*11.5%*	*12%*	*16%*	*19%*	*20%*	*17.1%*	*31.7%*	*30.0%*	*28.0%*	*26.3%*	*28.8%*	*25.0%*
FXHN Y/Y growth	***20.2%***	***23%***	***23%***	***20%***	***19%***	***21.1%***	***27.0%***	***29.0%***	***29.0%***	***27.0%***	***28.0%***	***25.0%***
Q/Q growth		*-9%*	*10%*	*6%*	*13%*		*0.4%*	*8.1%*	*4.1%*	*11.8%*		
Other Americas as a % of Gross Revenues	5.8%	6%	6%	6%	6%	5.9%	6.3%	6.4%	6.3%	6.5%	6.4%	6.8%
Foreign Exchange Impact on Gross Revenue												
Total Gross Revenue	350,018.0	90,234	96,428	102,346	113,828	402,836.0	109,394.5	116,415.2	122,346.1	134,447.2	482,603.0	564,364.8
Y/Y growth	***13.9%***	***12%***	***14%***	***16%***	***18%***	***15.1%***	***21.2%***	***20.7%***	***19.5%***	***18.1%***	***19.8%***	***16.9%***
Q/Q growth		***-6%***	***7%***	***6%***	***11%***		***-3.9%***	***6.4%***	***5.1%***	***9.9%***		
Y/Y Foreign Exchange Impact	(2,939.0)	(1,605)	543	1,339	1,331	1,608.0	3,826.7	1,299.8	(84.2)	615.8	5,658.1	119.6
	-7%	-16%	-21%	-15%	-5%	-8%	-20%	-20%	-20%	-20%	-20%	-20%
Cash Flow Hedging Impact	211.0	260	(112)	(207)	(68)	(127.0)	(765.3)	(260.0)	16.8	(123.2)	(1,131.6)	(23.9)
FXHN Gross Revs. (based on year ago period)	352,746.0	91,579	95,997	101,214	112,565	401,355.0	106,333.1	115,375.4	122,413.5	133,954.6	478,076.6	564,245.2
Y/Y growth (ex-FX & Hedging)	***14.8%***	***14%***	***13%***	***15%***	***17%***	***14.7%***	***18.2%***	***19.5%***	***19.4%***	***17.6%***	***18.6%***	***16.6%***
BPS of Y/Y growth impacted by FX & Hedging	(97.6)	(177.2)	37.2	126.0	128.5	35.4	305.2	121.7	17.6	50.3	116.1	29.8
FX & Hedging Neutral Gross Revs. (based on last quarter)		90,995	94,919	101,635	114,153		108,963.4	116,316.1	122,346.1	134,447.2		
Q/Q growth (ex-FX & Hedging)		***-6%***	***5%***	***5%***	***12%***		***-4.3%***	***5.6%***	***4.9%***	***9.9%***		
BPS of Q/Q growth impacted by FX & Hedging		(80.8)	136.8	86.0	(31.8)		43.6	82.9	23.4	(1.5)		

$ millions except where noted	2024A	3/25A	6/25A	9/25A	12/25A	2025A	3/26E	6/26E	9/26E	12/26E	2026E	2027E
Foreign Exchange Impact on EMEA Gross Revenue (excl. Hedging)												
Total Gross EMEA Revenue	102,127.0	25,923	28,262	29,911	33,056	117,152.0	32,918.0	34,262.0	34,969.7	38,436.2	140,585.8	161,003.1
Y/Y growth	***12.2%***	***9%***	***14%***	***17%***	***17%***	***14.7%***	***27.0%***	***21.2%***	***16.9%***	***16.3%***	***20.0%***	***14.5%***
Q/Q growth		***-8%***	***9%***	***6%***	***11%***		***-0.4%***	***4.1%***	***2.1%***	***9.9%***		
Y/Y Foreign Exchange Impact	38.0	(724)	780	1,261	1,438	2,755.0	3,236.1	1,365.0	332.7	587.0	5,520.9	32.4
FXHN Gross Revs. (based on year ago period)	102,089	26,647	27,482	28,650	31,618	114,397.0	29,681.8	32,897.0	34,636.9	37,849.1	135,064.9	160,970.8
Y/Y growth (excl. FX & Hedging)	***12.1%***	***12%***	***11%***	***12%***	***12%***	***12.0%***	***14.5%***	***16.4%***	***15.8%***	***14.5%***	***15.3%***	***14.5%***
BPS of Y/Y growth impacted by FX & Hedging	4.2	(304.4)	316.0	495.1	510.2	269.8	1,248.4	483.0	111.2	177.6	471.3	2.3
Y/Y growth in FX	*0.0%*	*-2.7%*	*2.8%*	*4.4%*	*4.5%*	*2.4%*	*10.9%*	*4.1%*	*1.0%*	*1.6%*	*4.1%*	*0.0%*
Foreign Exchange Impact on APAC Gross Revenue (excl. Hedging)												
Total Gross APAC Revenue	56,815.0	14,854	16,480	17,819	18,527	67,680.0	18,910.2	20,974.6	22,453.9	23,419.8	85,758.5	103,854.9
Y/Y growth	***10.3%***	***12%***	***19%***	***22%***	***22%***	***19.1%***	***27.3%***	***27.3%***	***26.0%***	***26.4%***	***26.7%***	***21.1%***
Q/Q growth		***-2%***	***11%***	***8%***	***4%***		***2.1%***	***10.9%***	***7.1%***	***4.3%***		
Y/Y Foreign Exchange Impact	(1,369.0)	(381)	115	128	(176)	(314.0)	342.7	(119.8)	(354.4)	75.8	(55.8)	87.2
FXHN Gross Revs. (based on year ago period)	58,184	15,235	16,365	17,691	18,703	67,994.0	18,567.5	21,094.4	22,808.3	23,344.0	85,814.2	103,767.7
Y/Y growth (excl. FX & Hedging)	***12.9%***	***15%***	***18%***	***22%***	***23%***	***19.7%***	***25.0%***	***28.0%***	***28.0%***	***26.0%***	***26.8%***	***21.0%***
BPS of Y/Y growth impacted by FX & Hedging	(265.8)	(286.7)	83.2	88.0	(116.1)	(55.3)	230.7	(72.7)	(198.9)	40.9	(8.2)	10.2
Y/Y growth in FX	*-2.4%*	*-2.5%*	*0.7%*	*0.7%*	*-0.9%*	*-0.5%*	*1.8%*	*-0.6%*	*-1.6%*	*0.3%*	*-0.1%*	*0.1%*
Foreign Exchange Impact on Other Americas Gross Revenue (excl. Hedging)												
Total Gross Other Americas Revenue	20,418.0	5,233.0	5,735.0	6,065.0	6,869.0	23,902.0	6,893.8	7,452.7	7,761.4	8,676.6	30,784.5	38,480.6
Y/Y growth	***11.5%***	***12%***	***16%***	***19%***	***20%***	***17.1%***	***31.7%***	***30.0%***	***28.0%***	***26.3%***	***28.8%***	***25.0%***
Q/Q growth		***-9%***	***10%***	***6%***	***13%***		***0.4%***	***8.1%***	***4.1%***	***11.8%***		
Y/Y Foreign Exchange Impact	(1,608.0)	(500.0)	(352.0)	(50.0)	69.0	(833.0)	247.9	54.5	(62.5)	(47.0)	192.9	-
FXHN Gross Revs. (based on year ago period)	22,026	5,733	6,087	6,115	6,800	24,735.0	6,645.9	7,398.2	7,823.9	8,723.6	30,591.5	38,480.6
Y/Y growth (excl. FX & Hedging)	***20.2%***	***23%***	***23%***	***20%***	***19%***	***21.1%***	***27.0%***	***29.0%***	***29.0%***	***27.0%***	***28.0%***	***25.0%***
BPS of Y/Y growth impacted by FX & Hedging	(877.7)	(1,074.6)	(712.8)	(98.2)	120.3	(408.0)	473.8	95.1	(103.0)	(68.5)	80.7	-
Y/Y growth in FX	*-7.3%*	*-8.7%*	*-5.8%*		*1.0%*	*-3.4%*	*3.7%*	*0.7%*	*-0.8%*	*-0.5%*	*0.6%*	*0.0%*
Headcount												
Ending Full-time Headcount (ex-Motorola)	183,323	185,719	187,103	190,167	190,820	190,820	192,520	194,520	197,520	198,520	198,520	205,468
Y/Y growth	*0.4%*	*2.7%*	*4.2%*	*4.9%*	*4.1%*	*4.1%*	*3.7%*	*4.0%*	*3.9%*	*4.0%*	*4.0%*	*3.5%*
Q/Q growth		*1.3%*	*0.7%*	*1.6%*	*0.3%*		*0.9%*	*1.0%*	*1.5%*	*0.5%*		
Absolute Sequential Change in Headcount	821	2,396	1,384	3,064	653	7,497	1,700	2,000	3,000	1,000	7,700	6,948

Source: J.P. Morgan estimates, Company data.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Figure 6: Alphabet Segment Reporting

$ in millions, except per share amounts	2024A	3/25A	6/25A	9/25A	12/25A	2025A	3/26E	6/26E	9/26E	12/26E	2026E	2027E
ALPHABET SEGMENT SUMMARY												
Revenues:												
Google Services	304,930	77,264	82,543	87,052	95,862	342,721	90,000	94,116	97,893	107,939	389,948	432,822
Google Cloud	43,229	12,260	13,624	15,157	17,664	58,705	19,690	22,045	23,932	26,086	91,752	128,486
Other Bets	1,648	450	373	344	370	1,537	470	514	505	545	2,035	3,080
Hedging gains (losses)	211	260	(112)	(207)	(68)	(127)	(765)	(260)	17	(123)	(1,132)	(24)
Total Revenues	**350,018**	**90,234**	**96,428**	**102,346**	**113,828**	**402,836**	**109,394**	**116,415**	**122,346**	**134,447**	**482,603**	**564,365**
Total Revenues ex-hedge	349,807	89,974	96,540	102,553	113,896	402,963	110,160	116,675	122,329	134,570	483,735	564,389
GAAP Operating Income (Loss):												
Google Services	121,263	32,682	33,063	33,527	40,132	139,404	40,050	40,658	42,094	45,874	168,676	190,442
Google Cloud	6,112	2,177	2,826	3,594	5,313	13,910	4,726	5,952	6,462	7,304	24,443	34,691
Other Bets	(4,444)	(1,226)	(1,246)	(1,426)	(3,617)	(7,515)	(1,406)	(1,443)	(1,478)	(1,797)	(6,124)	(5,662)
Corporate costs, unallocated	(10,541)	(3,027)	(3,372)	(4,467)	(5,894)	(16,760)	(5,172)	(4,752)	(4,754)	(4,968)	(19,645)	(21,922)
Total Income from Operations	**112,390**	**30,606**	**31,271**	**31,228**	**35,934**	**129,039**	**38,198**	**40,415**	**42,323**	**46,413**	**167,350**	**197,549**
Margins:												
Google Services	39.8%	42.3%	40.1%	38.5%	41.9%	40.7%	44.5%	43.2%	43.0%	42.5%	43.3%	44.0%
Google Cloud	14.1%	17.8%	20.7%	23.7%	30.1%	23.7%	24.0%	27.0%	27.0%	28.0%	26.6%	27.0%
Other Bets	-269.7%	-272.4%	-334.0%	-414.5%	-977.6%	-488.9%	-298.8%	-280.5%	-292.9%	-329.7%	-301.0%	-183.8%
Corporate costs, unallocated (% of total revenues)	-3.0%	-3.4%	-3.5%	-4.4%	-5.2%	-4.2%	-4.7%	-4.1%	-3.9%	-3.7%	-4.1%	-3.9%
Total Income from Operations	**32.1%**	**33.9%**	**32.4%**	**30.5%**	**31.6%**	**32.0%**	**34.9%**	**34.7%**	**34.6%**	**34.5%**	**34.7%**	**35.0%**
SEGMENT ANALYSIS												
GOOGLE SERVICES												
Revenue	304,930	77,264	82,543	87,052	95,862	342,721	90,000	94,116	97,893	107,939	389,948	432,822
TAC	54,900	13,748	14,705	14,876	16,597	59,926	15,657	16,129	16,406	18,123	66,315	72,336
Net Revenue	250,030	63,516	67,838	72,176	79,265	282,795	74,343	77,987	81,487	89,816	323,632	360,486
GAAP Costs & Opex	128,767	30,834	34,775	38,649	39,133	143,391	34,293	37,329	39,393	43,942	154,956	170,044
GAAP Operating Income	121,263	32,682	33,063	33,527	40,132	139,404	40,050	40,658	42,094	45,874	168,676	190,442
OI&E	3,493	1,465	1,459	2,199	881	6,004	980	873	733	667	3,252	2,076
Income before taxes	124,756	34,147	34,522	35,726	41,013	145,408	41,029	41,531	42,827	46,541	171,928	192,517
Income Tax	19,697	7,249	5,737	9,008	4,662	26,656	6,268	6,606	6,889	7,533	27,296	31,940
GAAP Net Income	105,059	26,898	28,785	26,718	36,351	118,752	34,761	34,924	35,938	39,008	144,632	160,577
GAAP EPS	$8.44	$2.19	$2.36	$2.19	$2.97	$9.71	$2.84	$2.84	$2.92	$3.16	$11.76	$13.03
D&A	10,490					13,709					22,617	30,298
SBC	16,161					17,205					19,107	21,208
Adj. EBITDA	147,914					170,317					210,400	241,948
Costs & Expenses As % of Revenue												
TAC	18.0%	17.8%	17.8%	17.1%	17.3%	17.5%	17.4%	17.1%	16.8%	16.8%	17.0%	16.7%
GAAP Costs & Opex	42.2%	39.9%	42.1%	44.4%	40.8%	41.8%	38.1%	39.7%	40.2%	40.7%	39.7%	39.3%
D&A	3.4%					4.0%					5.8%	7.0%
SBC	5.3%					5.0%					4.9%	4.9%
Margins			excl. EC fine	**42.5%**		**41.7%**						
GAAP Operating Income	**39.8%**	**42.3%**	**40.1%**	**38.5%**	**41.9%**	**40.7%**	**44.5%**	**43.2%**	**43.0%**	**42.5%**	**43.3%**	**44.0%**
GOOGLE CLOUD							13%	10%	6%	2%	7%	13%
Revenue	43,229	12,260	13,624	15,157	17,664	58,705	19,690	22,045	23,932	26,086	91,752	128,486
GAAP Costs & Opex	37,117	10,083	10,798	11,563	12,351	44,795	14,964	16,093	17,470	18,782	67,309	93,795
GAAP Operating Income	6,112	2,177	2,826	3,594	5,313	13,910	4,726	5,952	6,462	7,304	24,443	34,691
OI&E	0	0	0	0	0	0	0	0	0	0	0	0
Income before taxes	6,112	2,177	2,826	3,594	5,313	13,910	4,726	5,952	6,462	7,304	24,443	34,691
Income Tax	0	0	0	0	0	0	0	0	0	0	0	0
GAAP Net Income	6,112	2,177	2,826	3,594	5,313	13,910	4,726	5,952	6,462	7,304	24,443	34,691
GAAP EPS	$0.49	$0.18	$0.23	$0.29	$0.43	$1.14	$0.39	$0.48	$0.52	$0.59	$1.99	$2.81
D&A	4,409					6,927					13,396	22,614
SBC	5,879					6,927					8,166	9,444
Adj. EBITDA	16,401					27,764					46,005	66,749
Costs & Expenses As % of Revenue												
GAAP Costs & Opex	85.9%	82.2%	79.3%	76.3%	69.9%	76.3%	76.0%	73.0%	73.0%	72.0%	73.4%	73.0%
D&A	10%					12%					15%	18%
SBC	14%					12%					9%	7%
Margins												
GAAP Operating Income	**14.1%**	**17.8%**	**20.7%**	**23.7%**	**30.1%**	**23.7%**	**24.0%**	**27.0%**	**27.0%**	**28.0%**	**26.6%**	**27.0%**
OTHER BETS												
Revenue	1,648	450	373	344	370	1,537	470	514	505	545	2,035	3,080
GAAP Costs & Opex	6,092	1,676	1,619	1,770	3,987	9,052	1,876	1,957	1,983	2,343	8,159	8,742
Non-GAAP Costs & Opex	4,935					7,731					6,635	7,184
GAAP Operating Income	**(4,444)**	**(1,226)**	**(1,246)**	**(1,426)**	**(3,617)**	**(7,515)**	**(1,406)**	**(1,443)**	**(1,478)**	**(1,797)**	**(6,124)**	**(5,662)**
OI&E	3,932	9,718	1,203	10,560	2,302	23,783	0	0	0	0	0	0
Income before taxes	(512)	8,492	(43)	9,134	(1,315)	16,268	(1,406)	(1,443)	(1,478)	(1,797)	(6,124)	(5,662)
Income Tax	0	0	0	0	0	0	0	0	0	0	0	0
GAAP Net Income	(512)	8,492	(43)	9,134	(1,315)	16,268	(1,406)	(1,443)	(1,478)	(1,797)	(6,124)	(5,662)
GAAP EPS	($0.04)	$0.69	($0.00)	$0.75	($0.11)	$1.33	($0.11)	($0.12)	($0.12)	($0.15)	($0.50)	($0.46)
D&A	412					500					620	720
SBC	745					821					904	838
Adj. EBITDA	(3,287)					(6,194)					(4,601)	(4,104)
Costs & Expenses As % of Revenue												
GAAP Costs & Opex	370%	372%	434%	515%	1078%	589%	399%	380%	393%	430%	401%	284%
Margins												
GAAP Operating Income	-269.7%	-272.4%	-334.0%	-414.5%	-977.6%	-488.9%	-298.8%	-280.5%	-292.9%	-329.7%	-301.0%	-183.8%
RECONCILING ITEMS												
Hedging gains (losses)	211	260	(112)	(207)	(68)	(127)	(765)	(260)	17	(123)	(1,132)	(24)
Corporate costs, unallocated	**8,965**	**3,287**	**3,260**	**3,653**	**5,826**	**16,026**	**4,406**	**4,492**	**4,771**	**4,845**	**18,514**	**21,898**
One-time items	1,787	0	0	607	0	607	0	0	0	0	0	0
GAAP Operating Income	(10,541)	(3,027)	(3,372)	(4,467)	(5,894)	(16,760)	(5,172)	(4,752)	(4,754)	(4,968)	(19,645)	(21,922)
OI&E	0	0	0	0	0	0	0	0	0	0	0	0
Income before taxes	(10,541)	(3,027)	(3,372)	(4,467)	(5,894)	(16,760)	(5,172)	(4,752)	(4,754)	(4,968)	(19,645)	(21,922)
Income Tax	0	0	0	0	0	0	0	0	0	0	0	0
GAAP Net Income	(10,541)	(3,027)	(3,372)	(4,467)	(5,894)	(16,760)	(5,172)	(4,752)	(4,754)	(4,968)	(19,645)	(21,922)
GAAP EPS	($0.85)	($0.25)	($0.28)	($0.37)	($0.48)	($1.37)	($0.42)	($0.39)	($0.39)	($0.40)	($1.60)	($1.78)
As % of Alphabet Total Revenue ex-hedge												
Corporate costs, unallocated	2.56%	3.65%	3.38%	3.56%	5.12%	3.98%	4.00%	3.85%	3.90%	3.60%	3.83%	3.88%

Source: J.P. Morgan estimates, Company data.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Investment Thesis, Valuation and Risks

Alphabet *(Overweight; Price Target: $395.00)*

Investment Thesis
We believe Alphabet's fundamentals are solid and the company will remain both a driver of and primary beneficiary of an increasingly digital economy & advances in Generative AI. Google remains focused on innovation, and we continue to believe there is a healthy runway across Search and YouTube ads as AI drives higher ROI & TV dollars shift online. Non-Ad businesses, incl. Cloud & Subscriptions, have significant headroom, and companies within Other Bets, notably Waymo, provide option value. We remain confident in the company's ability to innovate around Generative AI, control costs, & generate solid top-line growth.

Valuation
Our December 2026 PT of $395, up from prior $385, is based on 29x our '27E GAAP EPS of $13.60, which ex-cash equates to ~28x our '27E GAAP EPS ex-Other Bets of $14.06. We believe Alphabet shares should trade at a premium to the S&P 500 as Alphabet is one of the few companies in the S&P 500 with DD % revenue and EPS growth off of a very large base and a strong 32%+ GAAP Operating Income margin.

Risks to Rating and Price Target
Downside risks include: 1) Competition ramping across AI, Ads, Cloud, and for engineering/tech talent; **2)** potential for a return to heavy investment spending and margin compression; **3)** regulatory scrutiny resulting in meaningful change to operations or limiting innovation; and **4)** growth slowing more than anticipated with the next major revenue drivers, including Cloud, Hardware, and Waymo, not materializing as expected.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Investment Thesis, Valuation and Risks

Alphabet Inc. *(Overweight; Price Target: $395.00)*

Investment Thesis
We believe Alphabet's fundamentals are solid and the company will remain both a driver of and primary beneficiary of an increasingly digital economy & advances in Generative AI. Google remains focused on innovation, and we continue to believe there is a healthy runway across Search and YouTube ads as AI drives higher ROI & TV dollars shift online. Non-Ad businesses, incl. Cloud & Subscriptions, have significant headroom, and companies within Other Bets, notably Waymo, provide option value. We remain confident in the company's ability to innovate around Generative AI, control costs, & generate solid top-line growth.

Valuation
Our December 2026 PT of $395, up from prior $385, is based on 29x our '27E GAAP EPS of $13.60, which ex-cash equates to ~28x our '27E GAAP EPS ex-Other Bets of $14.06. We believe Alphabet shares should trade at a premium to the S&P 500 as Alphabet is one of the few companies in the S&P 500 with DD % revenue and EPS growth off of a very large base and a strong 32%+ GAAP Operating Income margin.

Risks to Rating and Price Target
Downside risks include: 1) Competition ramping across AI, Ads, Cloud, and for engineering/tech talent; **2)** potential for a return to heavy investment spending and margin compression; **3)** regulatory scrutiny resulting in meaningful change to operations or limiting innovation; and **4)** growth slowing more than anticipated with the next major revenue drivers, incl. Cloud, Hardware, and Waymo, not materializing as expected.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Alphabet: Summary of Financials

Income Statement - Annual	FY25A	FY26E	FY27E	FY28E
Revenue	**402,836**	**482,603**	**564,365**	**-**
COGS	(162,535)	(193,340)	(229,294)	-
Gross profit	**240,301**	**289,263**	**335,071**	**-**
SG&A	(46,718)	(50,487)	(55,845)	-
Adj. EBITDA	**178,585**	**232,159**	**282,670**	**-**
D&A	(21,136)	(36,633)	(53,631)	-
Adj. EBIT	**157,449**	**195,527**	**229,039**	**-**
Net Interest	-	-	-	-
Adj. PBT	**187,236**	**198,779**	**231,115**	**-**
Tax	(26,656)	(27,296)	(31,940)	-
Minority Interest	-	-	-	-
Adj. Net Income	**134,971**	**168,309**	**195,435**	**-**
Reported EPS	10.81	11.66	13.60	-
Adj. EPS	**11.04**	**13.69**	**15.85**	**-**
DPS	**0.82**	**0.84**	**0.84**	**-**
Payout ratio	7.6%	7.2%	6.2%	-
Shares outstanding	12,230	12,295	12,327	-

Income Statement - Quarterly	1Q26E	2Q26E	3Q26E	4Q26E
Revenue	**109,394**	**116,415**	**122,346**	**134,447**
COGS	(43,731)	(46,516)	(49,142)	(53,951)
Gross profit	**65,663**	**69,900**	**73,205**	**80,496**
SG&A	(10,592)	(12,335)	(12,554)	(15,006)
Adj. EBITDA	**51,509**	**55,559**	**59,485**	**65,607**
D&A	(7,030)	(8,223)	(9,746)	(11,632)
Adj. EBIT	**44,479**	**47,335**	**49,739**	**53,974**
Net Interest	-	-	-	-
Adj. PBT	**45,458**	**48,208**	**50,472**	**54,641**
Tax	(6,268)	(6,606)	(6,889)	(7,533)
Minority Interest	-	-	-	-
Adj. Net Income	**38,519**	**40,828**	**42,730**	**46,232**
Reported EPS	2.69	2.82	2.94	3.21
Adj. EPS	**3.14**	**3.32**	**3.47**	**3.75**
DPS	**0.21**	**0.21**	**0.21**	**0.21**
Payout ratio	7.8%	7.4%	7.1%	6.6%
Shares outstanding	12,255	12,282	12,309	12,336

Balance Sheet & Cash Flow Statement	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	30,708	28,498	22,827	-
Accounts receivable	62,886	75,235	88,923	-
Inventories	0	0	0	-
Other current assets	112,444	85,647	74,196	-
Current assets	**206,038**	**189,379**	**185,946**	**-**
PP&E	246,597	390,732	546,213	-
LT investments	-	-	-	-
Other non current assets	142,646	145,836	149,372	-
Total assets	**595,281**	**725,947**	**881,531**	**-**
Short term borrowings	0	0	0	-
Payables	12,200	14,596	17,251	-
Other short term liabilities	90,545	108,325	128,034	-
Current liabilities	**102,745**	**122,921**	**145,285**	**-**
Long-term debt	46,547	46,547	43,547	-
Other long term liabilities	30,724	30,724	29,932	-
Total liabilities	**180,016**	**200,192**	**218,764**	**-**
Shareholders' equity	415,265	525,755	662,767	-
Minority interests	-	-	-	-
Total liabilities & equity	**595,281**	**725,947**	**881,531**	**-**
BVPS	**33.95**	**42.76**	**53.77**	**-**
y/y Growth	*30.0%*	*25.9%*	*25.7%*	-
Net debt/(cash)	15,839	18,049	20,720	-
Cash flow from operating activities	**164,713**	**209,550**	**253,604**	**-**
o/w Depreciation & amortization	21,136	36,633	53,631	-
o/w Changes in working capital	618	1,434	798	-
Cash flow from investing activities	**(120,291)**	**(150,768)**	**(194,112)**	**-**
o/w Capital expenditure	(91,447)	(180,768)	(209,112)	-
as % of sales	*22.7%*	*37.5%*	*37.1%*	-
Cash flow from financing activities	**(37,388)**	**(60,992)**	**(65,163)**	**-**
o/w Dividends paid	-	(10,328)	(10,354)	-
o/w Net debt issued/(repaid)	32,138	0	(3,000)	-
Net change in cash	**7,242**	**(2,210)**	**(5,671)**	**-**
Adj. Free cash flow to firm	**73,266**	**28,782**	**44,492**	**-**
y/y Growth	*0.7%*	*(60.7%)*	*54.6%*	-

Ratio Analysis	FY25A	FY26E	FY27E	FY28E
Gross margin	59.7%	59.9%	59.4%	-
EBITDA margin	44.3%	48.1%	50.1%	-
EBIT margin	39.1%	40.5%	40.6%	-
Net profit margin	33.5%	34.9%	34.6%	-
ROE	36.5%	35.8%	32.9%	-
ROA	25.8%	25.5%	24.3%	-
ROCE	33.9%	32.6%	30.9%	-
SG&A/Sales	11.6%	10.5%	9.9%	-
Net debt/equity	0.0	0.0	0.0	-
P/E (x)	30.2	24.4	21.0	-
P/BV (x)	9.8	7.8	6.2	-
EV/EBITDA (x)	22.9	17.6	14.5	-
Dividend Yield	0.2%	0.3%	0.3%	-
Sales/Assets (x)	0.8	0.7	0.7	-
Interest cover (x)	-	-	-	-
Operating leverage	109.2%	122.1%	101.2%	-
Revenue y/y Growth	15.1%	19.8%	16.9%	-
EBITDA y/y Growth	18.7%	30.0%	21.8%	-
Tax rate	14.2%	13.7%	13.8%	-
Adj. Net Income y/y Growth	15.6%	24.7%	16.1%	-
EPS y/y Growth	17.7%	24.0%	15.8%	-
DPS y/y Growth	38.9%	2.2%	0.0%	-

Source: Company reports and J.P. Morgan estimates.
Note: $ in millions (except per-share data).Fiscal year ends Dec. o/w - out of which

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Alphabet Inc.: Summary of Financials

Income Statement - Annual	FY25A	FY26E	FY27E	FY28E
Revenue	**402,836**	**482,603**	**564,365**	**-**
COGS	(162,535)	(193,340)	(229,294)	-
Gross profit	**240,301**	**289,263**	**335,071**	**-**
SG&A	(46,718)	(50,487)	(55,845)	-
Adj. EBITDA	**178,585**	**232,159**	**282,670**	**-**
D&A	(21,136)	(36,633)	(53,631)	-
Adj. EBIT	**157,449**	**195,527**	**229,039**	**-**
Net Interest	-	-	-	-
Adj. PBT	**187,236**	**198,779**	**231,115**	**-**
Tax	(26,656)	(27,296)	(31,940)	-
Minority Interest	-	-	-	-
Adj. Net Income	**134,971**	**168,309**	**195,435**	**-**
Reported EPS	10.81	11.66	13.60	-
Adj. EPS	**11.04**	**13.69**	**15.85**	**-**
DPS	**0.82**	**0.84**	**0.84**	**-**
Payout ratio	7.6%	7.2%	6.2%	-
Shares outstanding	12,230	12,295	12,327	-

Income Statement - Quarterly	1Q26E	2Q26E	3Q26E	4Q26E
Revenue	**109,394**	**116,415**	**122,346**	**134,447**
COGS	(43,731)	(46,516)	(49,142)	(53,951)
Gross profit	**65,663**	**69,900**	**73,205**	**80,496**
SG&A	(10,592)	(12,335)	(12,554)	(15,006)
Adj. EBITDA	**51,509**	**55,559**	**59,485**	**65,607**
D&A	(7,030)	(8,223)	(9,746)	(11,632)
Adj. EBIT	**44,479**	**47,335**	**49,739**	**53,974**
Net Interest	-	-	-	-
Adj. PBT	**45,458**	**48,208**	**50,472**	**54,641**
Tax	(6,268)	(6,606)	(6,889)	(7,533)
Minority Interest	-	-	-	-
Adj. Net Income	**38,519**	**40,828**	**42,730**	**46,232**
Reported EPS	2.69	2.82	2.94	3.21
Adj. EPS	**3.14**	**3.32**	**3.47**	**3.75**
DPS	**0.21**	**0.21**	**0.21**	**0.21**
Payout ratio	7.8%	7.4%	7.1%	6.6%
Shares outstanding	12,255	12,282	12,309	12,336

Balance Sheet & Cash Flow Statement	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	30,708	28,498	22,827	-
Accounts receivable	62,886	75,235	88,923	-
Inventories	0	0	0	-
Other current assets	112,444	85,647	74,196	-
Current assets	**206,038**	**189,379**	**185,946**	**-**
PP&E	246,597	390,732	546,213	-
LT investments	-	-	-	-
Other non current assets	142,646	145,836	149,372	-
Total assets	**595,281**	**725,947**	**881,531**	**-**
Short term borrowings	0	0	0	-
Payables	12,200	14,596	17,251	-
Other short term liabilities	90,545	108,325	128,034	-
Current liabilities	**102,745**	**122,921**	**145,285**	**-**
Long-term debt	46,547	46,547	43,547	-
Other long term liabilities	30,724	30,724	29,932	-
Total liabilities	**180,016**	**200,192**	**218,764**	**-**
Shareholders' equity	415,265	525,755	662,767	-
Minority interests	-	-	-	-
Total liabilities & equity	**595,281**	**725,947**	**881,531**	**-**
BVPS	**33.95**	**42.76**	**53.77**	**-**
y/y Growth	*30.0%*	*25.9%*	*25.7%*	-
Net debt/(cash)	15,839	18,049	20,720	-
Cash flow from operating activities	**164,713**	**209,550**	**253,604**	**-**
o/w Depreciation & amortization	21,136	36,633	53,631	-
o/w Changes in working capital	618	1,434	798	-
Cash flow from investing activities	**(120,291)**	**(150,768)**	**(194,112)**	**-**
o/w Capital expenditure	(91,447)	(180,768)	(209,112)	-
as % of sales	*22.7%*	*37.5%*	*37.1%*	-
Cash flow from financing activities	**(37,388)**	**(60,992)**	**(65,163)**	**-**
o/w Dividends paid	(10,050)	(10,328)	(10,354)	-
o/w Net debt issued/(repaid)	32,138	0	(3,000)	-
Net change in cash	**7,242**	**(2,210)**	**(5,671)**	**-**
Adj. Free cash flow to firm	**73,266**	**28,782**	**44,492**	**-**
y/y Growth	*0.7%*	*(60.7%)*	*54.6%*	-

Ratio Analysis	FY25A	FY26E	FY27E	FY28E
Gross margin	59.7%	59.9%	59.4%	-
EBITDA margin	44.3%	48.1%	50.1%	-
EBIT margin	39.1%	40.5%	40.6%	-
Net profit margin	33.5%	34.9%	34.6%	-
ROE	36.5%	35.8%	32.9%	-
ROA	25.8%	25.5%	24.3%	-
ROCE	33.9%	32.6%	30.9%	-
SG&A/Sales	11.6%	10.5%	9.9%	-
Net debt/equity	0.0	0.0	0.0	-
P/E (x)	30.2	24.3	21.0	-
P/BV (x)	9.8	7.8	6.2	-
EV/EBITDA (x)	22.9	17.6	14.5	-
Dividend Yield	0.2%	0.3%	0.3%	-
Sales/Assets (x)	0.8	0.7	0.7	-
Interest cover (x)	-	-	-	-
Operating leverage	109.2%	122.1%	101.2%	-
Revenue y/y Growth	15.1%	19.8%	16.9%	-
EBITDA y/y Growth	18.7%	30.0%	21.8%	-
Tax rate	14.2%	13.7%	13.8%	-
Adj. Net Income y/y Growth	15.6%	24.7%	16.1%	-
EPS y/y Growth	17.7%	24.0%	15.8%	-
DPS y/y Growth	38.9%	2.2%	0.0%	-

Source: Company reports and J.P. Morgan estimates.
Note: $ in millions (except per-share data).Fiscal year ends Dec. o/w - out of which

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Analyst Certification: The Research Analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst's personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst's analysis was made in good faith and that the views reflect the Research Analyst's own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker:** J.P. Morgan Securities LLC makes a market in the securities of Alphabet, Alphabet Inc. or related entities.
- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Alphabet, Alphabet Inc. or related entities.
- **Manager or Co-manager:** J.P. Morgan acted as manager or co-manager in a public offering of securities or financial instruments (as such term is defined in Directive 2014/65/EU) of/for Alphabet, Alphabet Inc. or related entities within the past 12 months.
- **Beneficial Ownership (1% or more):** J.P. Morgan beneficially owns 1% or more of a class of common equity securities of Alphabet, Alphabet Inc. or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Alphabet, Alphabet Inc. or related entities.
- **Client/Investment Banking:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as investment banking clients: Alphabet, Alphabet Inc. or related entities.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-investment-banking, securities-related: Alphabet, Alphabet Inc. or related entities.
- **Client/Non-Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-securities-related: Alphabet, Alphabet Inc. or related entities.
- **Investment Banking Compensation Received:** J.P. Morgan has received in the past 12 months compensation for investment banking services from Alphabet, Alphabet Inc. or related entities.
- **Potential Investment Banking Compensation:** J.P. Morgan expects to receive, or intends to seek, compensation for investment banking services in the next three months from Alphabet, Alphabet Inc. or related entities.
- **Non-Investment Banking Compensation Received:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from Alphabet, Alphabet Inc. or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Alphabet, Alphabet Inc. or related entities, if any.

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

Alphabet (GOOG, GOOG US) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Sep 28, 2004. All share prices are as of market close on the previous business day.

Date	Rating	Price ($)	Price Target ($)
26-Apr-23	OW	104.61	121
26-Jul-23	OW	122.79	150
13-Dec-23	OW	133.64	160
31-Jan-24	OW	153.05	165
26-Apr-24	OW	157.95	200
24-Jul-24	OW	183.60	208
30-Oct-24	OW	171.14	212
18-Dec-24	OW	197.12	232
05-Feb-25	OW	207.71	220
08-Apr-25	OW	149.24	180
25-Apr-25	OW	161.47	195
11-Jul-25	OW	178.70	200
24-Jul-25	OW	191.51	232
03-Sep-25	OW	211.99	260
27-Oct-25	OW	260.51	300
30-Oct-25	OW	275.17	340
12-Dec-25	OW	313.70	385

Alphabet Inc. (GOOGL, GOOGL US) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jul 17, 2015. All share prices are as of market close on the previous business day.

Date	Rating	Price ($)	Price Target ($)
26-Apr-23	OW	103.85	121
26-Jul-23	OW	122.21	150
13-Dec-23	OW	132.52	160
31-Jan-24	OW	151.46	165
26-Apr-24	OW	156.00	200
24-Jul-24	OW	181.79	208
30-Oct-24	OW	169.68	212
18-Dec-24	OW	195.42	232
05-Feb-25	OW	206.38	220
08-Apr-25	OW	146.75	180
25-Apr-25	OW	159.28	195
11-Jul-25	OW	177.62	200
24-Jul-25	OW	190.23	232
03-Sep-25	OW	211.35	260
27-Oct-25	OW	259.92	300
30-Oct-25	OW	274.57	340
12-Dec-25	OW	312.43	385

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period.
J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:
J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, https://www.jpmorganmarkets.com.

Coverage Universe: Anmuth, Doug : Airbnb (ABNB), Alphabet (GOOG), Alphabet Inc. (GOOGL), Amazon.com (AMZN), Booking Holdings (BKNG), Chewy (CHWY), Criteo (CRTO), DoorDash (DASH), Expedia Group, Inc. (EXPE), Fiverr (FVRR), Instacart (CART), Lyft

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

(LYFT), Meta Platforms Inc (META), Netflix Inc (NFLX), Pattern Group Inc. (PTRN), Peloton (PTON), Pinterest (PINS), Reddit (RDDT), Snap Inc. (SNAP), Spotify (SPOT), StubHub (STUB), TripAdvisor, Inc. (TRIP), Uber Technologies (UBER), WEBTOON (WBTN), ZipRecruiter (ZIP), eBay, Inc (EBAY), trivago (TRVG)

J.P. Morgan Equity Research Ratings Distribution, as of January 01, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	51%	37%	12%
IB clients**	47%	47%	35%
JPMS Equity Research Coverage*	49%	38%	13%
IB clients**	72%	70%	53%

***Please note that the percentages may not add to 100% because of rounding.**
****Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.**
For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at http://www.jpmorganmarkets.com, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, http://www.jpmorganmarkets.com. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:
A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of http://www.jpmorganmarkets.com where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to UK MIFID Research Unbundling exemption for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see https://www.jpmorgan.com/disclosures/cryptoasset-disclosure.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit https://www.theocc.com/components/docs/riskstoc.pdf for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:
Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting https://www.jpmm.com/research/disclosures. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: J.P. Morgan Australia - Research Independence Policy.

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC).

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: http://www.jpmipl.com. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Pranav Sata; pranav.d.sata@jpmchase.com; +912261573209. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit Terms and Conditions and Most Important Terms and Conditions (MITC). The annual Compliance audit report is available at http://www.jpmipl.com/#research.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission.

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: J.P. Morgan EMEA - Research Independence Policy.

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: https://www.jpmorgan.com/disclosures/email

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.

Doug Anmuth [AC]
(1-212) 622-6571
douglas.anmuth@jpmorgan.com

J.P.Morgan

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to msci.com/disclaimer

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised January 01, 2026.

**Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 05 Feb 2026 02:28 AM EST

Disseminated 05 Feb 2026 02:28 AM EST

This document is being provided for the exclusive use of YUBO SUI at THE CHINESE UNIVERSITY OF HONG KONG.